Logarithmic and Exponential Functions

ClipSAT Printable Practice Suite (clipsat.org) • Topic 4.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Logarithmic & Exponential Functions Practice Exercises**

Solve the exponential equation $2^{x+2} = 5^x$, giving x in terms of common logarithms.

Final Answer: $x = \frac{2\ln 2}{\ln 5 - \ln 2}$. Take natural logs of both sides:

$(x+2)\ln 2 = x\ln 5 \Rightarrow x\ln 2 + 2\ln 2 = x\ln 5 \Rightarrow x(\ln 5 - \ln 2) = 2\ln 2 \Rightarrow x = \frac{2\ln 2}{\ln 5 - \ln 2}$.

2 **[Foundational]** • **Target: Fundamental Laws of Logarithms**

Solve the exponential equation $2^{x+3} = 5^x$, giving x in terms of common logarithms.

Final Answer: $x = \frac{3\ln 2}{\ln 5 - \ln 2}$. Take natural logs of both sides:

$(x+3)\ln 2 = x\ln 5 \Rightarrow x\ln 2 + 3\ln 2 = x\ln 5 \Rightarrow x(\ln 5 - \ln 2) = 3\ln 2 \Rightarrow x = \frac{3\ln 2}{\ln 5 - \ln 2}$.

3 **[Foundational]** • **Target: Solving Logarithmic Equations**

Solve the exponential equation $2^{x+4} = 5^x$, giving x in terms of common logarithms.

Final Answer: $x = \frac{4\ln 2}{\ln 5 - \ln 2}$. Take natural logs of both sides:

$(x+4)\ln 2 = x\ln 5 \Rightarrow x\ln 2 + 4\ln 2 = x\ln 5 \Rightarrow x(\ln 5 - \ln 2) = 4\ln 2 \Rightarrow x = \frac{4\ln 2}{\ln 5 - \ln 2}$.

4 **[Foundational]** • **Target: Exponential Equations and Growth Models**

Solve the exponential equation $2^{x+5} = 5^x$, giving x in terms of common logarithms.

Final Answer: $x = \frac{5\ln 2}{\ln 5 - \ln 2}$. Take natural logs of both sides:

$(x+5)\ln 2 = x\ln 5 \Rightarrow x\ln 2 + 5\ln 2 = x\ln 5 \Rightarrow x(\ln 5 - \ln 2) = 5\ln 2 \Rightarrow x = \frac{5\ln 2}{\ln 5 - \ln 2}$.

5 **[Foundational]** • **Target: Reduction to Linear Form ($Y = mX + c$)**

Solve the exponential equation $2^{x+6} = 5^x$, giving x in terms of common logarithms.

Final Answer: $x = \frac{6\ln 2}{\ln 5 - \ln 2}$. Take natural logs of both sides:

$(x+6)\ln 2 = x\ln 5 \Rightarrow x\ln 2 + 6\ln 2 = x\ln 5 \Rightarrow x(\ln 5 - \ln 2) = 6\ln 2 \Rightarrow x = \frac{6\ln 2}{\ln 5 - \ln 2}$.

6 **[Foundational]** • **Target: Logarithmic & Exponential Functions Practice Exercises**

Solve the exponential equation $2^{x+7} = 5^x$, giving x in terms of common logarithms.

Final Answer: $x = \frac{7\ln 2}{\ln 5 - \ln 2}$. Take natural logs of both sides:

$(x+7)\ln 2 = x\ln 5 \Rightarrow x\ln 2 + 7\ln 2 = x\ln 5 \Rightarrow x(\ln 5 - \ln 2) = 7\ln 2 \Rightarrow x = \frac{7\ln 2}{\ln 5 - \ln 2}$.

© ClipSAT — clipsat.org | All Rights Reserved

7 [Foundational] • Target: Fundamental Laws of Logarithms

Solve the exponential equation $2^{x+8} = 5^x$, giving x in terms of common logarithms.

Final Answer: $x = \frac{8\ln 2}{\ln 5 - \ln 2}$. Take natural logs of both sides:

$(x+8)\ln 2 = x\ln 5 \Rightarrow x\ln 2 + 8\ln 2 = x\ln 5 \Rightarrow x(\ln 5 - \ln 2) = 8\ln 2 \Rightarrow x = \frac{8\ln 2}{\ln 5 - \ln 2}$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Solving Logarithmic Equations

The real numbers x and y satisfy the simultaneous equations: $\log_3(x) + 2\log_9(y-1) = 2$ $4^y \cdot 8^{-\frac{x}{3}} = 32$ Find the value of $x^2 + y^2$.

Final Answer: 25. Step 1: Simplify the logarithmic equation using the change-of-base formula. Recall that $\log_9(y-1) = \frac{\log_3(y-1)}{\log_3(9)} = \frac{\log_3(y-1)}{2}$. Substituting this gives:
$\log_3(x) + 2\left(\frac{\log_3(y-1)}{2}\right) = 2 \Rightarrow \log_3(x) + \log_3(y-1) = 2$ Applying the product rule of logarithms:
$\log_3[x(y-1)] = 2 \Rightarrow x(y-1) = 3^2 = 9 \quad - - -$ (Equation 1) Step 2: Simplify the exponential equation using laws of indices. Ex.

9 [Intermediate] • Target: Exponential Equations and Growth Models

The variables x and y, where $x > 0$, satisfy the relationship $y = ax + \frac{b}{x}$ where a and b are positive constants. When xy is plotted on the vertical axis against x^2 on the horizontal axis, a straight line is obtained which passes through the points $(2, 11)$ and $(5, 23)$. Given further that x and y satisfy the logarithmic equation $\log_3(y - ax) + \log_9(x^2y^2) = 1 + \log_3(x + 10)$ find the sum of all possible values of x.

Final Answer: $\frac{10}{3}$. We proceed in three main stages: **Stage 1: Straight-Line Reduction to determine a and b** Multiplying $y = ax + \frac{b}{x}$ by x yields: $xy = a(x^2) + b$ Letting $Y = xy$ and $X = x^2$, the equation is in the linear form $Y = aX + b$. The line passes through $(2, 11)$ and $(5, 23)$. The gradient is:
$a = \frac{23-11}{5-2} = \frac{12}{3} = 4$ Substituting into $Y = 4X + b$ using $(2, 11)$: $11 = 4(2) + b \Rightarrow b = 3$ Thus, $y = 4x + \frac{3}{x}$, which implies.

10 [Intermediate] • Target: Reduction to Linear Form ($Y = mX + c$)

Solve the logarithmic equation $\log_2(5x + 6) - \log_2(x) = 3$.

Final Answer: $x = 2$. 1. Apply the logarithm quotient law $\log_a(A) - \log_a(B) = \log_a\left(\frac{A}{B}\right)$: $\log_2\left(\frac{5x+6}{x}\right) = 3$ 2. Convert the equation from logarithmic to exponential form: $\frac{5x+6}{x} = 2^3 = 8$ 3. Multiply through by x (noting $x > 0$): $5x + 6 = 8x \Rightarrow 3x = 6 \Rightarrow x = 2$ 4. Checking the domain: Both $5(2) + 6 = 16 > 0$ and $2 > 0$, so $x = 2$ is a valid solution. Common misconceptions: - Calculating $2^3 = 6$ instead of 8 yields.

© ClipSAT — clipsat.org | All Rights Reserved

11 [Intermediate] • Target: Logarithmic & Exponential Functions Practice Exercises

Solve the simultaneous system: $\log_2(x) - \log_2(y) = 2$ and $x - 3y = 3$. What is the value of x? Show your complete algebraic work.

Final Answer: 12. From the first equation: $\log_2(x/y) = 2 \Rightarrow x/y = 2^2 = 4 \Rightarrow x = 4y$. Substitute into second: $4y - 3y = 3 \Rightarrow y = 3$. Thus $x = 4(3) = 12$.

12 [Intermediate] • Target: Fundamental Laws of Logarithms

Solve the simultaneous system: $\log_2(x) - \log_2(y) = 2$ and $x - 3y = 4$. What is the value of x? Show your complete algebraic work.

Final Answer: 16. From the first equation: $\log_2(x/y) = 2 \Rightarrow x/y = 2^2 = 4 \Rightarrow x = 4y$. Substitute into second: $4y - 3y = 4 \Rightarrow y = 4$. Thus $x = 4(4) = 16$.

13 [Intermediate] • Target: Solving Logarithmic Equations

Solve the simultaneous system: $\log_2(x) - \log_2(y) = 2$ and $x - 3y = 5$. What is the value of x? Show your complete algebraic work.

Final Answer: 20. From the first equation: $\log_2(x/y) = 2 \Rightarrow x/y = 2^2 = 4 \Rightarrow x = 4y$. Substitute into second: $4y - 3y = 5 \Rightarrow y = 5$. Thus $x = 4(5) = 20$.

14 [Intermediate] • Target: Exponential Equations and Growth Models

Solve the simultaneous system: $\log_2(x) - \log_2(y) = 2$ and $x - 3y = 6$. What is the value of x? Show your complete algebraic work.

Final Answer: 24. From the first equation: $\log_2(x/y) = 2 \Rightarrow x/y = 2^2 = 4 \Rightarrow x = 4y$. Substitute into second: $4y - 3y = 6 \Rightarrow y = 6$. Thus $x = 4(6) = 24$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Reduction to Linear Form ($Y = mX + c$) • Cross-Lesson: 5.01 Trigonometric Functions and Circular Measure

Solve the simultaneous system: $\log_2(x) - \log_2(y) = 2$ and $x - 3y = 7$. What is the value of x? Show your complete algebraic work.

Final Answer: 28. From the first equation: $\log_2(x/y) = 2 \Rightarrow x/y = 2^2 = 4 \Rightarrow x = 4y$. Substitute into second: $4y - 3y = 7 \Rightarrow y = 7$. Thus $x = 4(7) = 28$.

© ClipSAT — clipsat.org | All Rights Reserved

16 [Challenging] • Target: Logarithmic & Exponential Functions Practice Exercises • Cross-Lesson: 6.01 Permutations and Combinations

Solve the simultaneous system: $\log_2(x) - \log_2(y) = 2$ and $x - 3y = 8$. What is the value of x? Show your complete algebraic work.

Final Answer: 32. From the first equation: $\log_2(x/y) = 2 \Rightarrow x/y = 2^2 = 4 \Rightarrow x = 4y$. Substitute into second: $4y - 3y = 8 \Rightarrow y = 8$. Thus $x = 4(8) = 32$.

17 [Challenging] • Target: Fundamental Laws of Logarithms • Cross-Lesson: 7.01 Differentiation and Rates of Change

Solve the simultaneous system: $\log_2(x) - \log_2(y) = 2$ and $x - 3y = 9$. What is the value of x? Show your complete algebraic work.

Final Answer: 36. From the first equation: $\log_2(x/y) = 2 \Rightarrow x/y = 2^2 = 4 \Rightarrow x = 4y$. Substitute into second: $4y - 3y = 9 \Rightarrow y = 9$. Thus $x = 4(9) = 36$.

18 [Challenging] • Target: Solving Logarithmic Equations • Cross-Lesson: 8.01 Integration and Kinematics

Variables x and y satisfy the equation $y = Ax^k$, where A and k are constants, with $x > 0$. The graph of $\ln y$ against $\ln x$ is a straight line passing through the points $(\ln 2, \ln 72)$ and $(\ln 6, \ln 8)$. Find the value of x when $y = 18$.

Final Answer: 4. Step 1: Reduce the non-linear relationship to linear form. Taking the natural logarithm of both sides of $y = Ax^k$: $\ln y = \ln(Ax^k) = \ln A + k\ln x$ This is in the linear form $Y = mX + c$, where $Y = \ln y$, $X = \ln x$, gradient $m = k$, and vertical intercept $c = \ln A$. Step 2: Determine the gradient k. Using the given points $(\ln 2, \ln 72)$ and $(\ln 6, \ln 8)$.

19 [Challenging] • Target: Exponential Equations and Growth Models • Cross-Lesson: 1.01 Functions, Modulus and Quadratics

The variables x and y satisfy the relationship $y = Ax^n$, where A and n are constants. When $\log_2 y$ is plotted against $\log_2 x$, a straight line is obtained which passes through the points $(1, 5)$ and $(4, -1)$. What is the value of y when $x = 4$? Show your complete algebraic work.

Final Answer: 8. 1. Express the non-linear relationship in linear form $Y = mX + c$: Taking base-2 logarithms of both sides of $y = Ax^n$: $\log_2 y = \log_2(Ax^n) = n\log_2 x + \log_2 A$ Letting $Y = \log_2 y$ and $X = \log_2 x$, the equation becomes: $Y = nX + \log_2 A$ where the gradient is $m = n$ and the vertical intercept is $c = \log_2 A$. 2. Find the equation of the straight line: Using the given points $(X_1, Y_1) = (1, 5)$ and $(X_2, Y_2) = (4, -1)$: $m = \frac{-1-5}{4-1} = \frac{-6}{3} = -2$ Usin.

© ClipSAT — clipsat.org | All Rights Reserved

20 [Challenging] • Target: Reduction to Linear Form ($Y = mX + c$) • Cross-Lesson: 2.01 Simultaneous Equations and Inequalities

The variables x and y satisfy the equation $y = Ab^x$, where A and b are positive constants. When $\log_2 y$ is plotted against x, a straight line is obtained that passes through the points $(2, 7)$ and $(5, 13)$. Find the value of y when $x = 4$.

Final Answer: 2048. To reduce the non-linear relationship $y = Ab^x$ to linear form, take base-2 logarithms of both sides: $\log_2 y = \log_2 (Ab^x) = \log_2 A + x\log_2 b$ Setting $Y = \log_2 y$ and $X = x$, the equation represents a straight line $Y = mX + c$, where the gradient is $m = \log_2 b$ and the vertical intercept is $c = \log_2 A$. 1. Find the gradient m: $m = \frac{13-7}{5-2} = \frac{6}{3} = 2$ 2. Find the vertical intercept c using the point $(2, 7)$: $7 = 2(2) + c \Rightarrow c = 3$ Thus, t.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Logarithmic & Exponential Functions Practice Exercises

Solve for \(x \): \(\log_{3}(x) + \log_{3}(4) = \log_{3}(16) \).

a. **4** ✓ **b.** 6

c. 8 **d.** 2

Using log laws: \(\log_{3}(4x) = \log_{3}(16) \implies 4x = 16 \implies x = 4 \).

22 [Foundational] • Target: Logarithmic & Exponential Functions Practice Exercises

Solve for \(x \): \(\log_{4}(x) + \log_{4}(4) = \log_{4}(36) \).

a. 3 **b.** 18

c. 11 **d.** **9** ✓

Using log laws: \(\log_{4}(4x) = \log_{4}(36) \implies 4x = 36 \implies x = 9 \).

23 [Foundational] • Target: Logarithmic & Exponential Functions Practice Exercises

Solve for \(x \): \(\log_{5}(x) + \log_{5}(4) = \log_{5}(64) \).

a. 4 **b.** 32

c. 18 **d.** **16** ✓

Using log laws: \(\log_{5}(4x) = \log_{5}(64) \implies 4x = 64 \implies x = 16 \).

© ClipSAT — clipsat.org | All Rights Reserved

24 [Foundational] • Target: Logarithmic & Exponential Functions Practice Exercises

Solve for x: $\log_{6}(x) + \log_{6}(4) = \log_{6}(100)$.

a. **25** ✓

b. 27

c. 5

d. 50

Using log laws: $\log_{6}(4x) = \log_{6}(100) \implies 4x = 100 \implies x = 25$.

25 [Foundational] • Target: Logarithmic & Exponential Functions Practice Exercises

Solve for x: $\log_{7}(x) + \log_{7}(4) = \log_{7}(144)$.

a. 38

b. **36** ✓

c. 72

d. 6

Using log laws: $\log_{7}(4x) = \log_{7}(144) \implies 4x = 144 \implies x = 36$.

26 [Foundational] • Target: Logarithmic & Exponential Functions Practice Exercises

Solve for x: $\log_{8}(x) + \log_{8}(4) = \log_{8}(196)$.

a. 7

b. 98

c. **49** ✓

d. 51

Using log laws: $\log_{8}(4x) = \log_{8}(196) \implies 4x = 196 \implies x = 49$.

27 [Foundational] • Target: Logarithmic & Exponential Functions Practice Exercises

Solve for x: $\log_{9}(x) + \log_{9}(4) = \log_{9}(256)$.

a. 66

b. 128

c. **64** ✓

d. 8

Using log laws: $\log_{9}(4x) = \log_{9}(256) \implies 4x = 256 \implies x = 64$.

28 [Foundational] • Target: Logarithmic & Exponential Functions Practice Exercises

Solve for x: $\log_{10}(x) + \log_{10}(4) = \log_{10}(324)$.

a. 9

b. 162

c. 83

d. **81** ✓

Using log laws: $\log_{10}(4x) = \log_{10}(324) \implies 4x = 324 \implies x = 81$.

© ClipSAT — clipsat.org | All Rights Reserved

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Fundamental Laws of Logarithms**

Evaluate $\log_3(54) - \log_3(2)$ without a calculator.

a. 3 ✓

b. 2

c. 4

d. $\log_3(52)$

$\log_3(54/2) = \log_3(27) = 3$ since $3^3 = 27$.

30 **[Intermediate]** • **Target: Fundamental Laws of Logarithms**

Express $2\log_a(x) - 3\log_a(y) + \log_a(z)$ as a single logarithm.

a. $\log_a\left(\frac{x^2z}{y^3}\right)$ ✓

b. $\log_a\left(\frac{2xz}{3y}\right)$

c. $\log_a(x^2 - y^3 + z)$

d. $\log_a\left(\frac{x^2}{y^3z}\right)$

Power law: $\log_a(x^2) - \log_a(y^3) + \log_a(z) = \log_a\left(\frac{x^2z}{y^3}\right)$.

31 **[Intermediate]** • **Target: Fundamental Laws of Logarithms**

Given $\log_b(2) = p$ and $\log_b(3) = q$, express $\log_b(72)$ in terms of p and q.

a. $3p + 2q$ ✓

b. $2p + 3q$

c. $6pq$

d. $3p - 2q$

$72 = 2^3 \times 3^2 \Rightarrow \log_b(72) = 3\log_b(2) + 2\log_b(3) = 3p + 2q$.

© ClipSAT — clipsat.org | All Rights Reserved

32 [Intermediate] • Target: Solving Logarithmic Equations

Solve the equation $\log_2(x) + \log_2(x-2) = 3$ for $x > 2$.

a. $x = 4$ ✓

b. $x = 4$ and $x = -2$

c. $x = 3$

d. $x = 8$

$\log_2(x(x-2)) = 3 \Rightarrow x^2 - 2x = 2^3 = 8 \Rightarrow x^2 - 2x - 8 = 0 \Rightarrow (x-4)(x+2) = 0$. Since $x > 2$, $x = 4$.

33 [Intermediate] • Target: Solving Logarithmic Equations

Solve $\log_3(2x+1) - \log_3(x-1) = 2$.

a. $x = \frac{10}{7}$ ✓

b. $x = 2$

c. $x = \frac{7}{10}$

d. No real solution

$\log_3\left(\frac{2x+1}{x-1}\right) = 2 \Rightarrow \frac{2x+1}{x-1} = 9 \Rightarrow 2x + 1 = 9x - 9 \Rightarrow 7x = 10 \Rightarrow x = 10/7$.

34 [Intermediate] • Target: Solving Logarithmic Equations

Solve the base-change logarithmic equation $\log_4(x) + \log_2(x) = 6$.

a. $x = 16$ ✓

b. $x = 8$

c. $x = 64$

d. $x = 4$

$\log_4(x) = \frac{\log_2(x)}{\log_2(4)} = \frac{1}{2}\log_2(x)$. Thus $\frac{3}{2}\log_2(x) = 6 \Rightarrow \log_2(x) = 4 \Rightarrow x = 2^4 = 16$.

35 [Intermediate] • Target: Exponential Equations and Growth Models

Solve $3^{2x-1} = 5^x$, giving x in terms of natural logarithms.

a. $x = \frac{\ln 3}{2\ln 3 - \ln 5}$ ✓

b. $x = \frac{\ln 5}{\ln 3}$

c. $x = \frac{\ln 3}{\ln 5 - 2\ln 3}$

d. $x = \frac{2\ln 3}{\ln 5}$

Take ln: $(2x-1)\ln 3 = x\ln 5 \Rightarrow 2x\ln 3 - \ln 3 = x\ln 5 \Rightarrow x(2\ln 3 - \ln 5) = \ln 3 \Rightarrow x = \frac{\ln 3}{2\ln 3 - \ln 5}$.

© ClipSAT — clipsat.org | All Rights Reserved

36 [Intermediate] • Target: Exponential Equations and Growth Models

A bacterial culture grows according to $N(t) = 500e^{0.04t}$, where t is in hours. After how many hours will the population reach 2000?

a. $t = 25\ln 4 \approx 34.7$ hr ✓

b. $t = 50$ hr

c. $t = 20\ln 2 \approx 13.9$ hr

d. $t = 100$ hr

$500e^{0.04t} = 2000 \Rightarrow e^{0.04t} = 4 \Rightarrow 0.04t = \ln 4 \Rightarrow t = \frac{\ln 4}{0.04} = 25\ln 4$.

37 [Intermediate] • Target: Exponential Equations and Growth Models

Solve the simultaneous exponential system $2^x 4^y = 32$ and $3^{x-y} = 27$.

a. $x = 4, y = \frac{1}{2}$ ✓

b. $x = 3, y = 1$

c. $x = 5, y = 2$

d. $x = 2, y = 1$

$2^{x+2y} = 2^5 \Rightarrow x + 2y = 5$. Also $3^{x-y} = 3^3 \Rightarrow x - y = 3$. Subtracting gives $3y = 2 \Rightarrow y = 1/2, x = 4$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Reduction to Linear Form ($Y = mX + c$) • Cross-Lesson: 8.01 Integration and Kinematics

The variables x and y are related by $y = ax^b$. When $\ln y$ is plotted against $\ln x$, a straight line passes through $(0, 2)$ with slope 3. Find a and b.

a. $a = e^2, \quad b = 3$ ✓

b. $a = 2, \quad b = 3$

c. $a = 3, \quad b = 2$

d. $a = e^3, \quad b = e^2$

$\ln y = \ln a + b\ln x$. Slope $m = b = 3$. Intercept $c = \ln a = 2 \Rightarrow a = e^2$.

© ClipSAT — clipsat.org | All Rights Reserved

39 [Challenging] • Target: Reduction to Linear Form ($Y = mX + c$) • Cross-Lesson: 1.01 Functions, Modulus and Quadratics

The relation $y = Ab^x$ is linearized by plotting $\lg y$ against x. If the straight line has y-intercept 1 and gradient 0.3, find A and b.

a. $A = 10, \quad b = 10^{0.3}$ ✓

b. $A = 1, \quad b = 0.3$

c. $A = 10^{0.3}, \quad b = 10$

d. $A = 0.3, \quad b = 10$

$\lg y = \lg A + x \lg b$. Intercept $\lg A = 1 \Rightarrow A = 10$. Gradient $\lg b = 0.3 \Rightarrow b = 10^{0.3}$.

40 [Challenging] • Target: Reduction to Linear Form ($Y = mX + c$) • Cross-Lesson: 2.01 Simultaneous Equations and Inequalities

If a graph of xy against x^2 is a straight line with slope 4 and vertical intercept -5, express y in terms of x.

a. $y = 4x - \frac{5}{x}$ ✓

b. $y = 4x^2 - 5$

c. $y = 4 - \frac{5}{x^2}$

d. $y = \frac{4}{x} - 5$

$xy = 4(x^2) - 5 \Rightarrow y = 4x - \frac{5}{x}$.

41 [Challenging] • Target: Logarithmic & Exponential Functions Practice Exercises • Cross-Lesson: 3.01 Indices, Surds and Polynomials

Find the real solution to the equation $\log_3 x + \log_3 (x - 8) = 2$.

a. $x = 9$ ✓

b. $x = 9$ and $x = -1$

c. $x = 8.5$

d. $x = 5$

Apply the product rule of logarithms, $\log_a u + \log_a v = \log_a (uv)$: $\log_3 [x(x - 8)] = 2$ Convert the logarithmic equation to index (exponential) form: $x(x - 8) = 3^2$ $x^2 - 8x = 9$ $x^2 - 8x - 9 = 0$ Factor the quadratic: $(x - 9)(x + 1) = 0$ which gives potential solutions $x = 9$ or $x = -1$. Check the domain of the original logarithmic expression: for $\log_3 x$ and $\log_3 (x - 8)$ to be defined for real numbers, we must have $x > 0$ and $x - 8 > 0$, which requires $x > 8$. Thus, $x = -1$ is an extra.

© ClipSAT — clipsat.org | All Rights Reserved

42 [Challenging] • Target: Fundamental Laws of Logarithms • Cross-Lesson: 5.01 Trigonometric Functions and Circular Measure

The variables x and y satisfy the relationship $y = Ab^x$, where A and b are positive constants. The graph of $\ln y$ against x is a straight line passing through the points $(0, \ln 3)$ and $(4, \ln 48)$. What is the value of b?

a. 2 ✓

b. 4

c. 16

d. $\sqrt{2}$

Taking the natural logarithm of both sides of the equation $y = Ab^x$: $\ln y = \ln(Ab^x) = \ln A + x \ln b$ This is in linear form $Y = mX + c$, where $Y = \ln y$, $X = x$, the gradient is $m = \ln b$, and the vertical intercept is $c = \ln A$. From the point $(0, \ln 3)$, the intercept is $\ln A = \ln 3$, which gives $A = 3$. Using the second point $(4, \ln 48)$: $\ln 48 = \ln 3 + 4 \ln b$ Subtract $\ln 3$ from both sides and apply the quotient rule of logarithms.

43 [Challenging] • Target: Solving Logarithmic Equations • Cross-Lesson: 6.01 Permutations and Combinations

Solve $e^{2x} - 23e^x + 130 = 0$ for x. Give the larger solution in exact form.

a. $\ln 10$

b. $\ln 13$ ✓

c. $\ln 23$

d. $\ln 130$

$(e^x - 10)(e^x - 13) = 0 \Rightarrow e^x = 10$ or $e^x = 13 \Rightarrow x = \ln 10$ or $x = \ln 13$. Larger: $x = \ln 13$.

44 [Challenging] • Target: Exponential Equations and Growth Models • Cross-Lesson: 7.01 Differentiation and Rates of Change

Solve $e^{2x} - 21e^x + 110 = 0$ for x. Give the larger solution in exact form.

a. $\ln 11$ ✓

b. $\ln 110$

c. $\ln 21$

d. $\ln 10$

$(e^x - 10)(e^x - 11) = 0 \Rightarrow e^x = 10$ or $e^x = 11 \Rightarrow x = \ln 10$ or $x = \ln 11$. Larger: $x = \ln 11$.

45 [Challenging] • Target: Reduction to Linear Form ($Y = mX + c$) • Cross-Lesson: 8.01 Integration and Kinematics

Solve $e^{2x} - 28e^x + 187 = 0$ for x. Give the larger solution in exact form.

a. $\ln 187$

b. $\ln 17$ ✓

c. $\ln 11$

d. $\ln 28$

$(e^x - 11)(e^x - 17) = 0 \Rightarrow e^x = 11$ or $e^x = 17 \Rightarrow x = \ln 11$ or $x = \ln 17$. Larger: $x = \ln 17$.

© ClipSAT — clipsat.org | All Rights Reserved